

Tessera — Mode A & Mode B

How the two-mode observation architecture functions today

Tessera runs **one detector substrate in two modes**, chosen per-emitter by a `validity_class` gate. **Mode A** is the always-on layer — it *ranks* shards by evidence and *abstains* when it cannot make a calibrated claim, with no false-discovery-rate promise. **Mode B** is a narrow, revocable upgrade that earns an **FDR $\leq q$** guarantee by building a valid *spatial* null from a concurrent control twin (`d = treatment - control`), which cancels the common-mode the temporal null cannot on nonstationary GPU telemetry.

■ Mode A — evidence / ranking +
abstain (default)

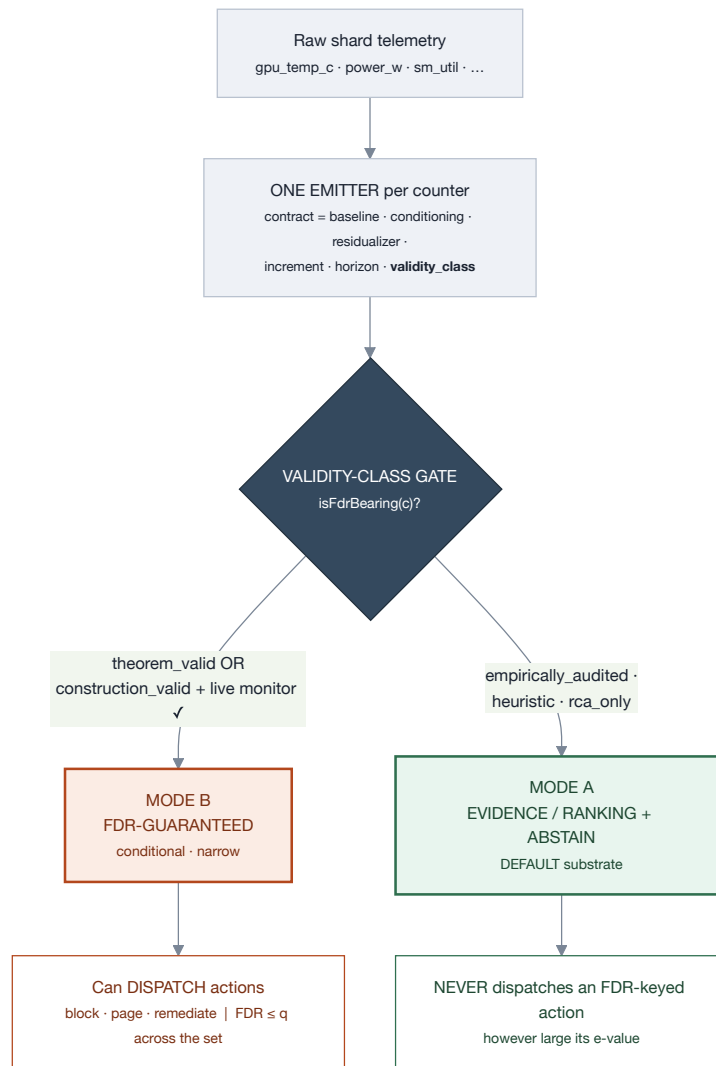
■ Mode B — FDR-guaranteed
(conditional, narrow)

■ Gate /
decision

Source: `tools/emitter-contract.ts · tools/mode-b-loop.ts · tools/contrast.ts · tools/control-triad.ts` | ADRs 0019-0022 | branch three-walls-prototypes | 2026-06-28

1 The big picture — one substrate, two modes, gated per-emitter

Every detector is an **emitter** (a whole contract). The gate alone decides its mode.



INTUITION

Mode A is the default — every emitter starts here. **Mode B is opt-in and revocable**: an emitter has to *earn* its way in and gets kicked back the moment its null stops being trustworthy. The gate (`assertFdrEligible`) *throws* rather than let an ineligible emitter silently acquire a guarantee — the same hard-fail discipline as the ≥ 56 -day baseline guard.

Mode A

ence, and **abstain** rather than bluff.



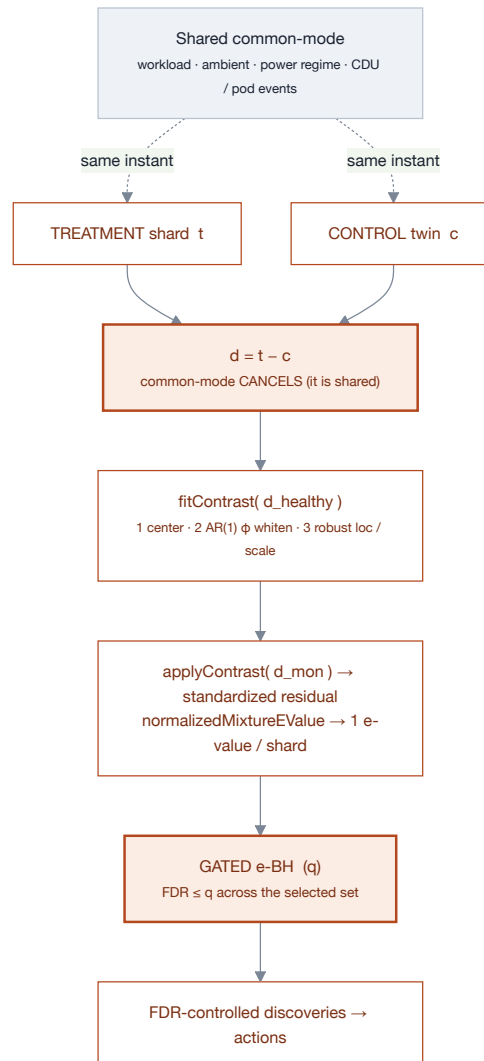
SHARD / EMITTER	EVIDENCE	MODE A BEHAVIOR
shard-417	 strong	surfaced top of the ranking
shard-031	 moderate	surfaced, lower
shard-902	 weak	below the fold
gpu_temp_c @ 1 Hz		ABSTAINS — cannot certify a valid null → says nothing

INTUITION

Mode A is **honest about uncertainty**: it surfaces and ranks, and when it cannot make a calibrated claim it *abstains* instead of guessing. `gpu_temp_c` at 1 Hz is the canonical abstainer — near-unit-root physics ($\phi \approx 0.992$) means there is no valid temporal null, so it stays in Mode A permanently. That abstention is *intrinsic and correct*, not a fixable artifact (confirmed against a genuine 2-month 1 Hz baseline).

3 Mode B FDR guarantee via a *spatial* null

A guarantee needs a valid null. On drifting telemetry, build it in **space**, not time.



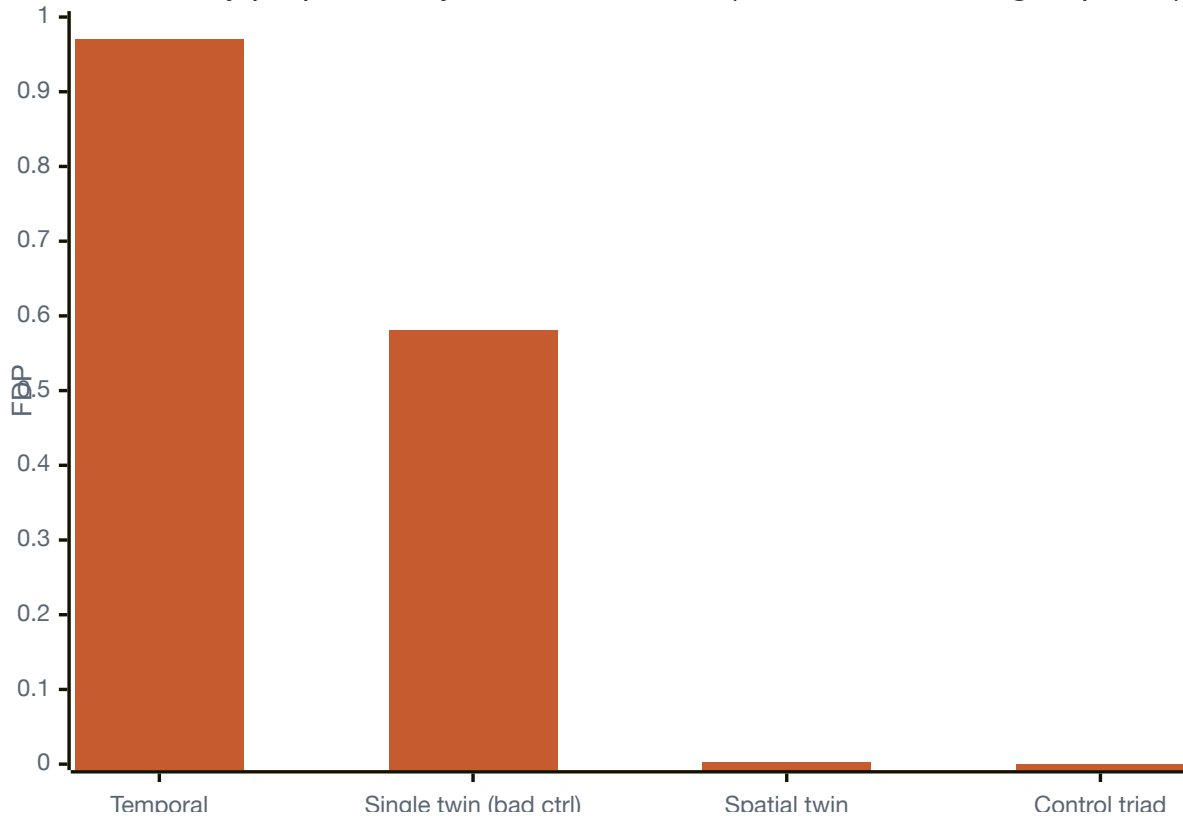
INTUITION

The **contrast** $d = t - c$ is the whole trick. Treatment and its twin feel the *same* workload, ambient temperature and power regime at the *same* instant, so subtracting them **cancels the common-mode exactly** — leaving a clean null the drifting temporal baseline could never give. Fit on a healthy window (center → whiten → standardize), apply to the monitoring window, turn each shard into one e-value, and run e-BH for $FDR \leq q$.

4 Why spatial, not temporal — the data

Same detector, same $q = 0.1$. Only the **null construction** changes.

False-discovery proportion by null construction (lower is better; target $q = 0.1$)

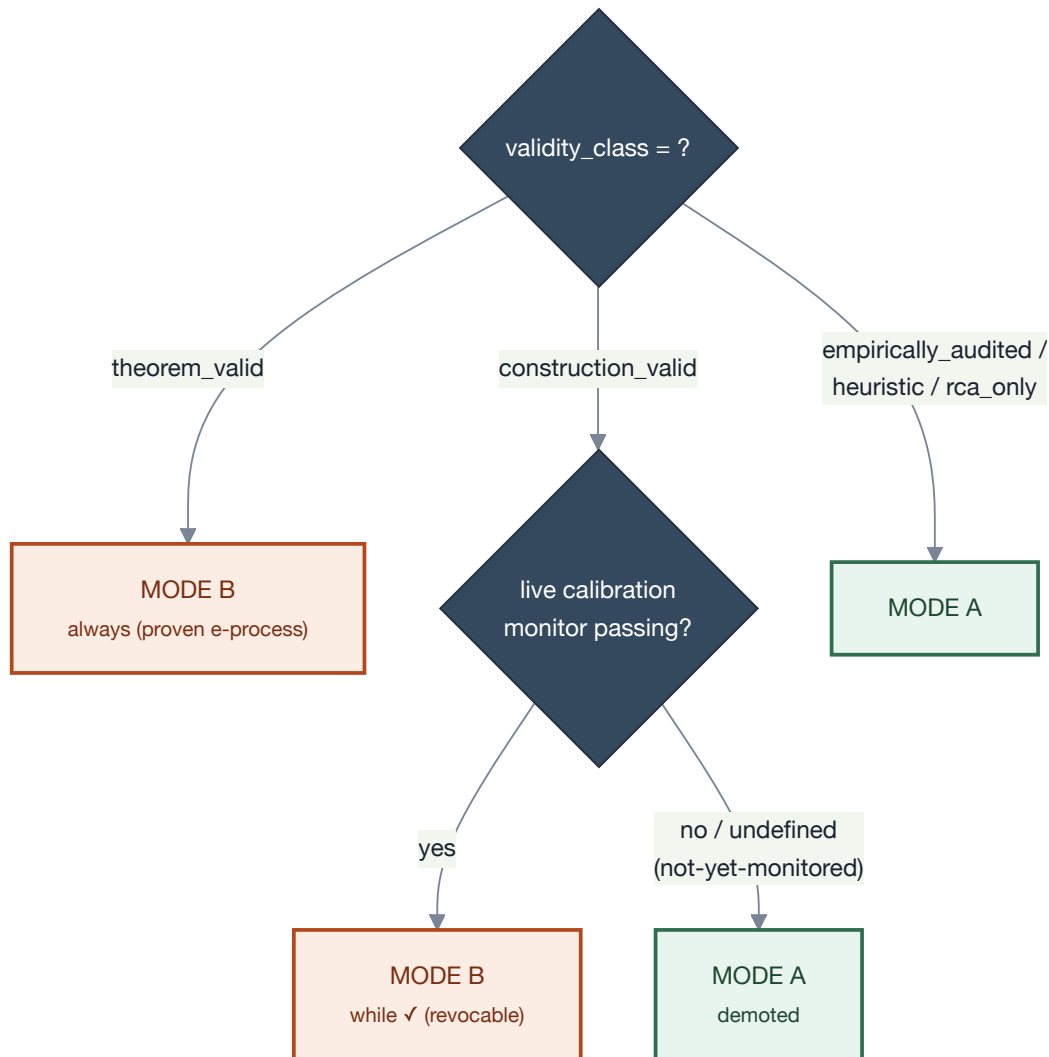


FDP at 1 Hz scale. Temporal over-selects catastrophically; a single twin fails when the control itself is contaminated; the matched spatial twin and the control triad sit at or below target.

Temporal compares a shard against its own past — drift contaminates it → $FDP \approx 0.97$. **Spatial** compares a shard against its twin at the same instant — common-mode cancels → $FDP \approx 0$. The **single twin** has one weakness (a contaminated *control* makes t-c false-fire, $FDP \approx 0.58$); the **triad** (§7) closes it.

5 The gate as a decision tree — what makes an emitter *Mode B right now*

Validity is a property of the live emitter, re-evaluated continuously.

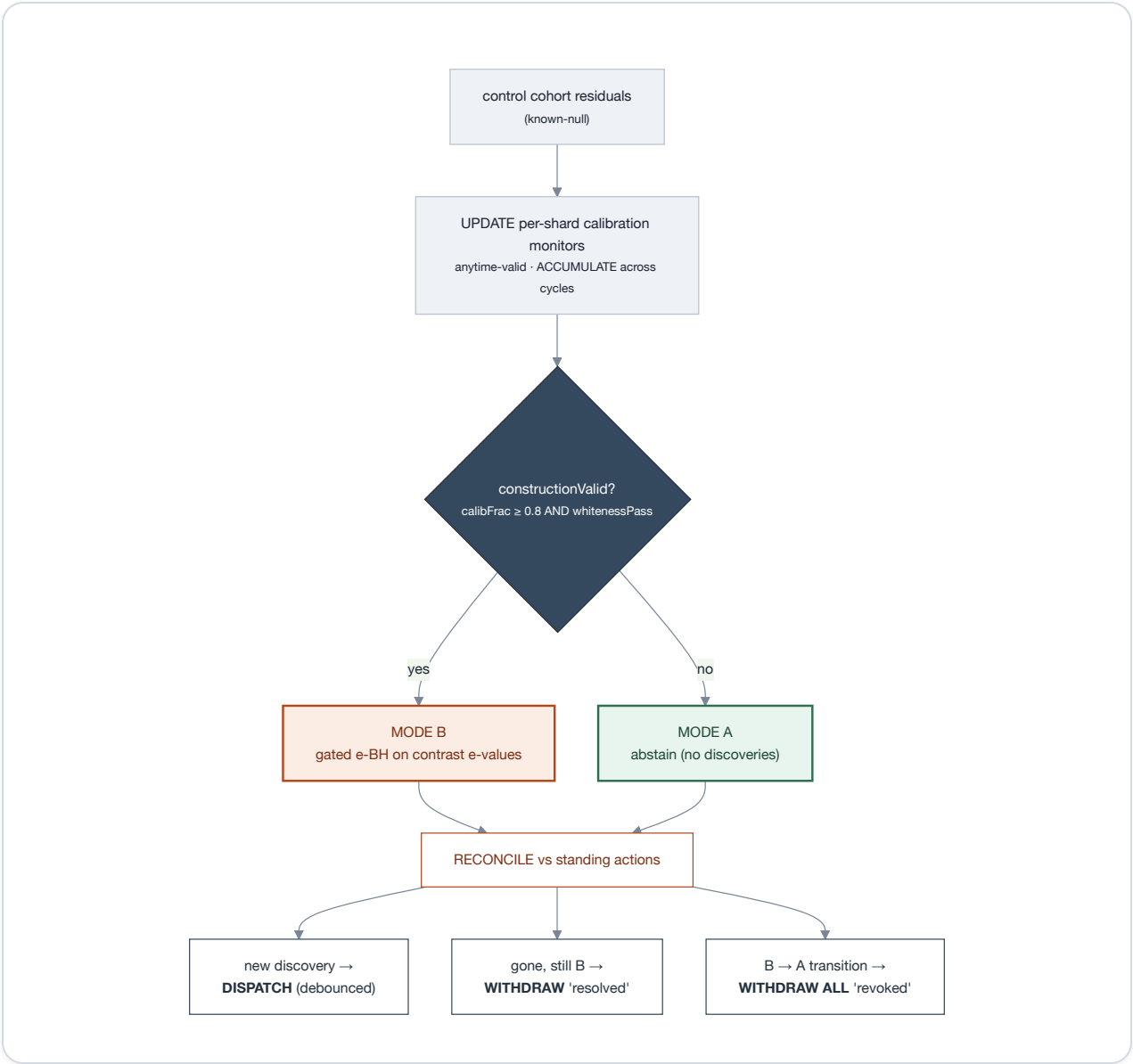


INTUITION

Two load-bearing rules: (1) `construction_valid` is **not a static stamp** — it is bound to a live monitor and requires `calibrationMonitorPassing == true` every cycle (undefined counts as failing — "a class without a live monitor manufactures false confidence"). (2) **The unit of validity is the emitter, not the counter** — the same counter can be Mode B under one contract and Mode A under another.

6 The always-on loop — how B and A behave over time

`mode-b-loop.ts` : a promotion / demotion machine, run every cycle, per emitter.



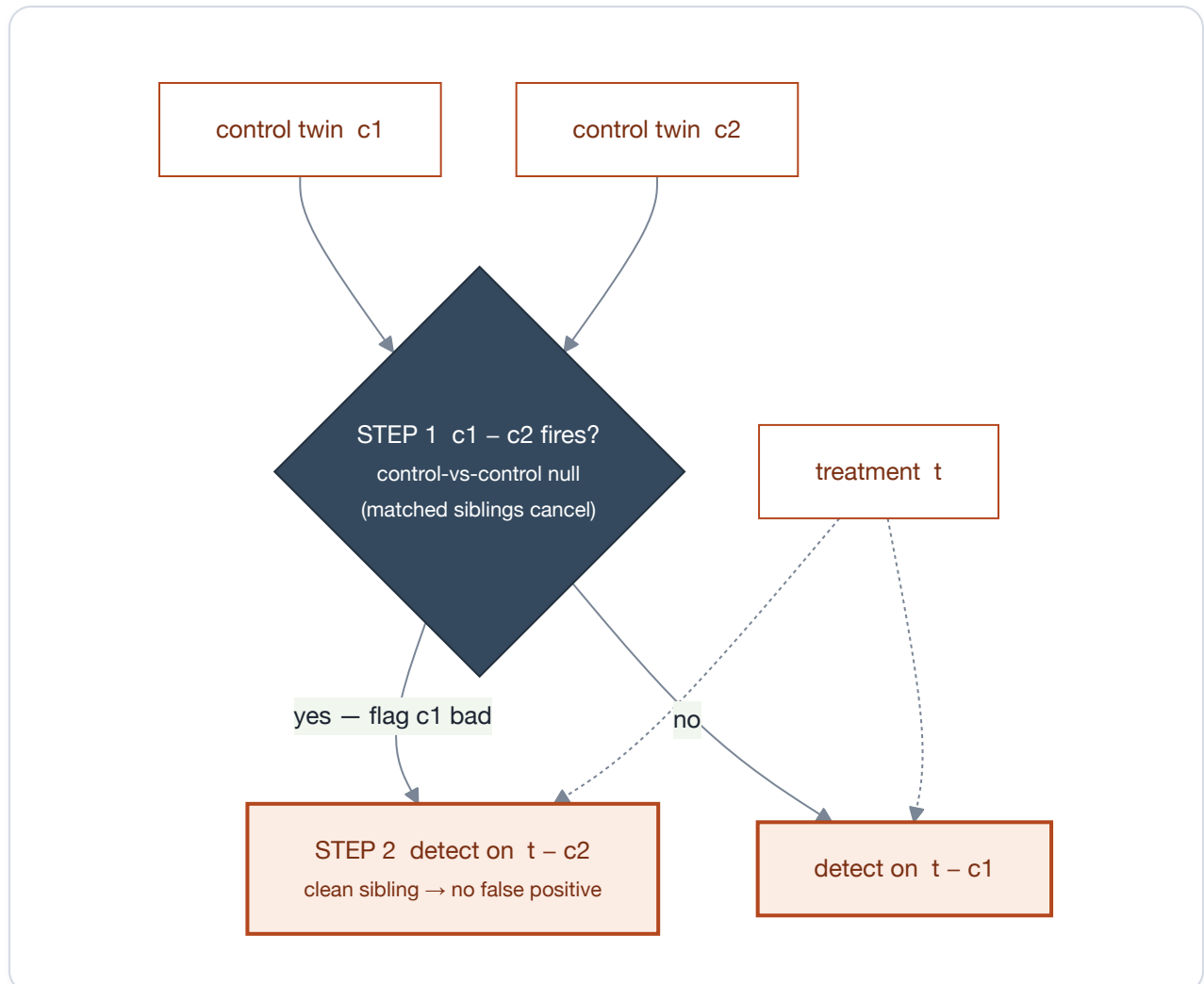
INVARIANT	MEANING
No action without a live guarantee	a Mode-A emitter never dispatches, however large its e-value
Revocation withdraws	guarantee breaks → all that emitter's standing actions are pulled
Debounce	a still-standing discovery fires once, not every cycle
Resolution withdraws	a discovery that disappears (while still Mode B) → 'resolved'

INTUITION

An emitter **rides into Mode B** when its construction holds, dispatches FDR-controlled actions, and **falls back to Mode A** — dropping *all* its actions — the instant the monitor or whiteness check says its null is no longer trustworthy. An action keyed to a bounded error rate must never outlive the guarantee it was keyed to.

7 The control triad (ADR 0022) — hardening the spatial null

A second twin lets the construction self-check when a *control* goes bad.

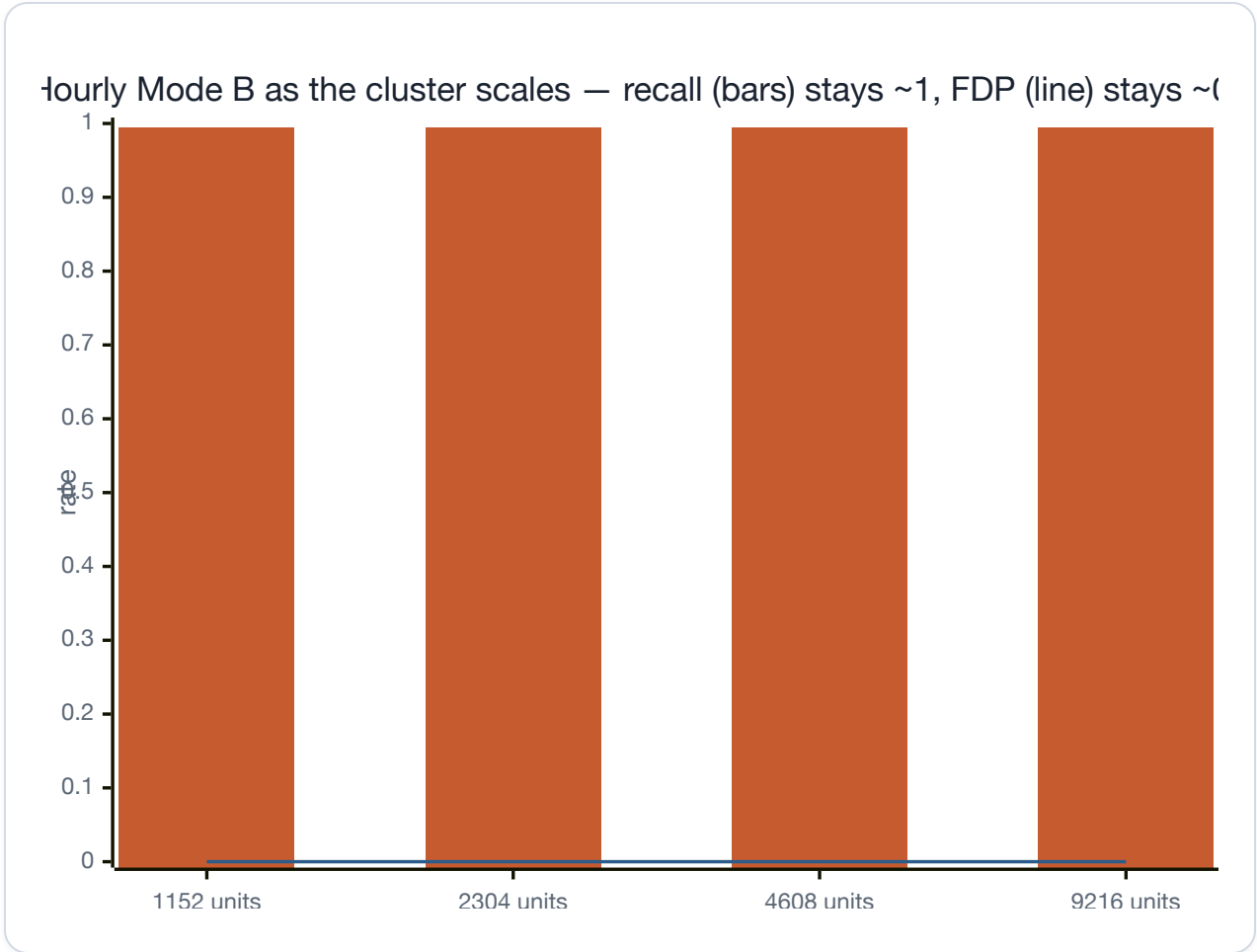


INTUITION

A single twin has one blind spot: if the *control* is contaminated, $t - c$ fires even though the treatment is healthy (a sign-blind false positive, FDP ≈ 0.58). The triad adds a second twin so $c1 - c2$ becomes a clean control-vs-control reference that flags the bad control, and detection routes to the clean sibling $t - c2 \rightarrow$ FDP back to **0.000** at recall 1.000. *Note:* `applyTriadRouting` is in-memory only today; wiring it into the streaming / 1 Hz-at-scale path is the top remaining item.

8 It holds at full cluster scale

Hourly Mode B, validated to **9,216 observation units** (4,608 treatment + 4,608 control).



Bars = recall (≈ 0.995). Line = FDP (≈ 0.000) across all four rack tiers. Cadence-dependent caveat: at 1 Hz, `power_w` certifies Mode B (FDP 0.000, recall 1.000) while `gpu_temp_c` correctly abstains to Mode A.

SETTING	SCALE	FDP	RECALL	NOTES
Hourly, full cluster	9,216 units	0.000	→0.995	every rack tier
1 Hz, 2-month baseline, <code>power_w</code>	9,216 units	0.000	1.000	certifies Mode B
1 Hz, <code>gpu_temp_c</code>	—	—	—	abstains (Mode A) — intrinsic, correct
Triad under control contamination	1,728 shards	0.000	1.000	vs single-pair FDP ≈ 0.58

INTUITION

The two-mode guarantee is **scale-invariant where it applies** and **honestly silent where it does not**. Growing the cluster 8× does not erode FDR control; switching to a near-unit-root counter does not produce false confidence — it produces abstention.